A hazard potential yielding a one-sample matroid prophet guarantee above 2^{-43}

Research addendum, October 10, 2026

Abstract

We replace the dyadic hazard-band argument in the OpenAI matroid prophet construction by a one-step potential argument. It gives expected exit hazard greater than $5/16$. Together with a sharper use of its residual encoding, this yields a proposed universal guarantee greater than 2^{-43} in the original one-sample, full-seed almighty-adversary model, with $44n+26$ fair seed bits. The exact coefficient is approximately $1.04159 \cdot 2^{-43}$. This is an AI-assisted draft with a written proof, exact scalar certificates, and finite checks of the new conditional inequalities; independent review and formalization are pending. The coefficient exceeds Daniel's announced 2^{-47} headline, but that proof's assumptions and strongest derived constant remain unavailable. We make no priority or best-known-bound claim.

1 Statement and dependencies

Proposed theorem 1. *For every known finite labeled matroid M on n elements, there is a measurable online rule using one independent sample per element and $44n + 26$ initial fair bits, whose law does not depend on the value distributions, with the following guarantee. Let all sample and value coordinates S_e, V_e be mutually independent and nonnegative, with matching laws at each label; let the seed be independent of these coordinates. Assuming only $\mathbb{E}\,\mathrm{OPT}_M(V) < \infty$, for every measurable arrival permutation depending on all samples, values, and the full seed, the rule accepts an independent set at every prefix and earns expected reward at least $2^{-43}\mathbb{E}\,\mathrm{OPT}_M(V)$.*

We use the density paths, reverse computation, hybrid safety comparison, marked-pivot encoding, and order-uniform accounting of OpenAI's manuscript [1]. The earlier note [2] specifies the algorithm and its affine branch mixing and finite-offset rounding. We prove the new hazard estimate here and state every changed quantitative deduction. Source combinatorial lemmas remain explicit dependencies. No polynomial-time algorithm or bounded observation precision is asserted.

2 One-step logarithmic resource

Fix a pre-step reverse transcript while a nonloop d is alive. The remaining coordinates V have their independent guard-bit laws. Let $\mathcal{A}$ be the increasing event that this transition survives, and let $p = \mathbb{P}(\mathcal{A}^c) > 0$. The source transition queries coordinates predictably, at most once, and its queried positive answers alone form a certificate for $\mathcal{A}$ whenever the transition survives.

Let V' be the remaining domain after the complete transition, and let q be the failure probability of $\mathcal{A}$ when only V' is random and every queried coordinate is reset to zero. Then $p \le q \le 1$ and

$$\mathbb{E}\log(q/p) \ge 1 - p. \tag{1}$$

To verify this, during the transaction write W for the remaining domain, I for the queried positives, $f(W)$ for the failure probability with only W random, and $f_I(W)$ for the failure probability with I forced positive. The ratio $Z = f_I(W)/f(W)$ starts at one and belongs to $[0, 1]$. For the next predictably selected query, $W' = W \setminus \{e\}$, product conditioning gives

$$\mathbb{E}[Z - Z' \mid \text{transcript}] = Z\left(1 - \frac{f(W)}{f(W')}\right) \le \log f(W') - \log f(W).$$

At survival the final ratio is zero; otherwise it is at most one. Thus its expected net decrease is at least $1-p$. Sum the display over the finite query sequence to obtain (1). A stopped sequence can be padded by zero increments. This is the source log-resource calculation applied to one transaction, without its later hazard-band loss.

3 The next hazard and a telescoping potential

Write the current reverse map as $\mathcal{G}_k(c, S)$ and its output as S'. The source query domains are precisely the departing coordinates. The remaining domain V' is the union of the group restrictions to S'. Conditional on the complete post-step transcript these bits retain their product law. On survival let r be the next conditional exit probability. For every assignment c supported on V', monotonicity in time and state gives

$$\mathcal{G}_{k-1}(c, S') \subseteq \mathcal{G}_k(c, S).$$

The right side resets queried bits to zero, just as the definition of q requires. Therefore the next survival event is contained in the depleted current survival event, so $r \ge q$ on survival. Both input states are frozen in this comparison; earlier states are not recomputed after resetting bits. No arrival order is conditioned upon.

Since exit has probability p and $\log(q/p) \le \log(1/p)$ on exit, (1) yields

$$\mathbb{E}[\mathbf{1}_{\text{survive}} \log(r/p)] \ge 1 - p - p\log(1/p).$$

Use $\log(r/p) \le (r-p)/p$ to conclude

$$\mathbb{E}[\mathbf{1}_{\text{survive}} r] - p \ge p - p^2(2 + \log(1/p)). \tag{2}$$

At $p = 0$ the right side is interpreted as zero and the inequality is immediate; at $p = 1$ it is equality.

Define a potential equal to the current conditional hazard while d is alive and zero after exit. Every nonloop exits within the deterministic finite reverse horizon. Summing (2) gives, if P is the hazard at the actual exit and p_{start} the initial hazard,

$$\mathbb{E}[P(2 + \log(1/P))] \ge 1 + p_{\text{start}} \ge 1. \tag{3}$$

Here $\sum \mathbb{E}[\mathbf{1}_{\text{alive}} p] = 1$, and weighting any function of p by the exit indicator is equivalent in expectation to multiplying by $\mathbf{1}_{\text{alive}} p$. Exits at zero hazard have probability zero.

The function $f(x) = x(2 - \log x)$, with $f(0) = 0$, is increasing and concave on $[0, 1]$. Jensen gives $f(\mathbb{E}P) \ge \mathbb{E}f(P) \ge 1$. Furthermore

$$e^{6/5} > \sum_{j=0}^{4} \frac{(6/5)^j}{j!} = \frac{2059}{625} > \frac{16}{5},$$

so $f(5/16) < (5/16)(2 + 6/5) = 1$. We have proved

Lemma 1. *Conditional on any fixed filter and density masks, every nonloop in the source reverse process has expected exit hazard strictly greater than* $5/16$*. Equivalently, its total expected squared-exit mass is greater than* $5/16$*.*

The source hybrid comparison turns squared-exit mass into pre-parity safety. Subtract the baseline-birth rank loss $|D_h|/\kappa$, average fair parity, and then average the independent quarter-rate density mask. The resulting safe-count coefficient is

$$\mathbb{E}[\sigma_i \mid H] \geq \gamma |Y_i| - \frac{n_i}{8\kappa}, \qquad \gamma = \frac{5}{128}.$$

4 Sharper residual count and tail

For $q = \lfloor n/\kappa \rfloor$ and $S_d(n) = \sum_{j=0}^{d} \binom{n}{j}$, the first marked sequence in the source residual encoding has only q movable density generators; the second has at most $2q$ generator and witness occurrences. Consequently its existing marked-pivot lemma gives

$$|\mathfrak{R}_i| \leq S_q(n)^2 (4^q S_q(n))(4^{2q} S_{2q}(n)).$$

The standard binomial bounds used in the source imply

$$\log |\mathfrak{R}_i| \leq \frac{n}{\kappa}(5 \log \kappa + 5 + 4 \log 2) \leq \frac{n}{\kappa}(5a + 12) \log 2 \quad (\kappa = 2^a \geq 4),$$

using $\log 2 > 2/3$. As in the source, this family is fixed conditional on the filter, parity, and exterior masks, before any focal bits or the focal listing event are exposed.

The probability of sacrifice at a focal coordinate is $p_O = (1+3t)/4$. Retain it in the tail estimate. Since $\log(1+3t) \leq 3t$ and $\log 2 > 2/3$,

$$\log_2(1/p_O) \geq 2 - 5t =: \ell.$$

Choose

$$\kappa = 2^{12}, \quad t = 2^{-20}, \quad B = 2^8, \quad \delta = \frac{84}{\ell \kappa}.$$

The residual-family union bound for every $n \geq \kappa$ is at most $2^{-12n/\kappa} \leq 2^{-12}$. Thus the nominal-rank estimate has transfer error

$$\epsilon = \frac{1}{8\kappa} + \frac{84}{\ell \kappa} + 2^{-12}.$$

Unlisted groups and outcomes with nominal rank greater than n/κ use the same deterministic branches as in the earlier derivation.

5 Final coefficient and evidence

Set $\beta = (B-1)/B$ and mix the main branch with probability $p = 1 - t/4 = 1 - 2^{-22}$. The order-uniform accounting from the earlier note gives the slack

$$s = \gamma/2 - \epsilon - t\kappa - \frac{t\kappa + 1/4}{B-1} = \frac{4790554237}{1168228319232}.$$

The maximum branch cancels the negative largest-coordinate term because $(1-p)/4 = t/16 \geq pt\gamma$. Average eight shared phases $J = 0, \ldots, 7$, all visible to the adversary, to obtain

$$C = \frac{p\beta^2 ts}{8\kappa} = \frac{25618620910087559025}{21634518663060144894698947241 5744} > 2^{-43}.$$

For an elementary check, $p\beta^2 > 127/128$ and $s > (128/127)/256$, while $t/\kappa = 2^{-32}$. The minimum over orders stays inside the seed expectation, and the source glued-vector simulation transfers the coefficient losslessly to the prophet model. Predrawing the masks costs $44n$ bits, and the phase, branch, and parity require $3 + 22 + 1 = 26$ more bits.

Exact rational code checks all displayed constants. Finite reverse-process checks cover 170 matroid/weight instances, 4,916 conditional drift inequalities, and 3,153 global exit-hazard inequalities. Logarithms in those checks use rational intervals from an atanh series after dyadic reduction. The checks include the comparison $r \geq q$, positive certificates, and a negative control demonstrating that monotone hazards alone are insufficient. These finite checks support the written proof; they do not prove its universal assertion. Same-agent second-pass verification is not independent review.

The coefficient is about 16.6655 times the announced 2^{-47} headline [3]. The unavailable announcement's exact model and strongest derived coefficient remain unresolved. The most important new review target is the depleted-event comparison used in (2). The earlier scalar-template limitation still holds for its old exit coefficient; Lemma 1 changes that ingredient.

References

[1] OpenAI. *One Sample Suffices for Matroid Prophet Inequalities against an Almighty Adversary.* September 23, 2026 manuscript, pinned at `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. Manuscript. See the reverse-map and log-resource equations in the safety section, and the two marked sequences in the residual-family proof.

[2] *A finite-seed parameter refinement for one-sample matroid prophet inequalities.* Preparatory research note, October 10, 2026. Repository PR #2.

[3] Daniel (@firsttimelifer). Announcement of a 2^{-47} guarantee, October 8, 2026. Proof unavailable in this audit.